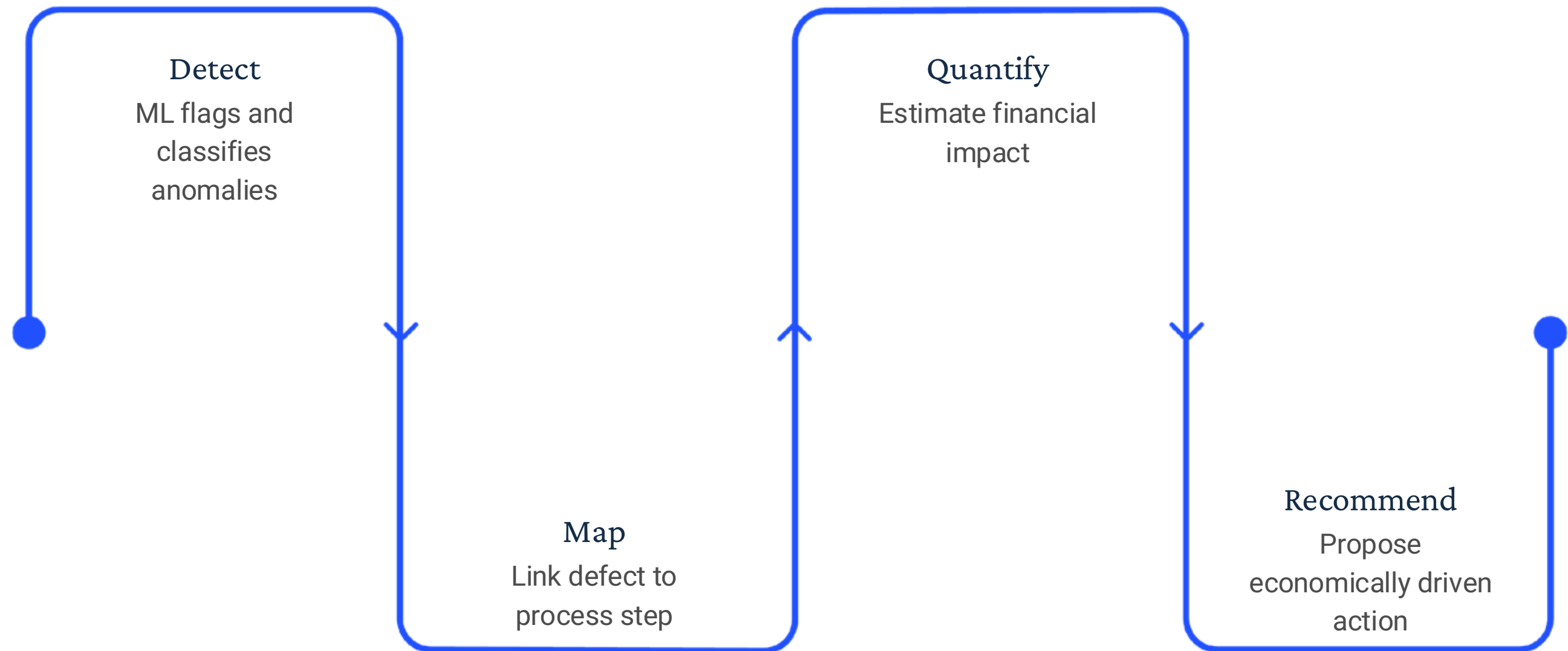


Strategic Shift: From Anomaly Detection to Decision Intelligence



Input: Wafer map anomalies report from production batch

Output: Prioritized repair recommendations for specific tools

Example Mapping: The "Donut" Defect Case Study

Defect: Donut

Likely Causes

- CMP pad wear (primary)
- Lithography redeposition (secondary)

Financial Impact

- Yield Impact: 30%
- Repair Cost: \$150,000
- Replacement: \$5.25M
- Downtime: \$1M/hr
- Frequency: Medium occurrence (18% of wafer defects)

[Source:](#) Semiconductor Digest (*A dirty Job: The Challenge of CMP*), EE Time (*CMP foundry expands services, capacity*)

OUTPUT

Example Executive Output: Actionable Decision Card

Defect: Edge_Ring Likely Process: Etch (Confidence: High)	Estimated Daily Loss: \$180k 30-Day Projection: \$5.4M Cost Basis: Tool-level throughput loss	Repair Cost: \$120k Replacement Cost: \$12M Frequency of Repair: Low Repair Lifetime: 12 months	ROI if Repaired: 3500% Recommended Action: Replace focus ring + recalibrate RF (Confidence: High)
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Source: Knowledge Sourcing Intelligence (*Semiconductor Dry Etch Systems Market - Strategic Insights and Forecast (2025-2030)*), Science Direct (*Edge Exclusion*), Semi (*When Minutes Matter: Time-Critical Logistics as a Competitive Edge*)